Class Activity # 4
Calculus Physics 2

ame: Key Date: _____

Group Members: _____ _____ _____

Solve the following problems together as a group, stay engaged in the conversation with the group!

➢ Use **Chapter 43 and 47 Lecture Notes** as a guide to answer the following questions.

1) As more resistors are added in parallel across a constant voltage source, the power supplied by the source

A

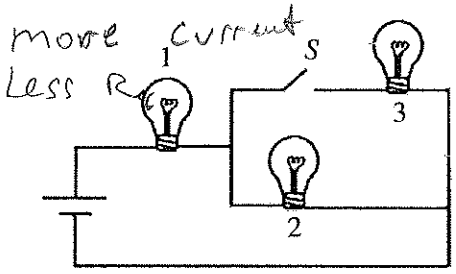
A) increases.
B) decreases.
C) does not change.

Lower resistance

2) The figure shows three identical lightbulbs connected to a battery having a constant voltage across its terminals. What happens to the brightness of lightbulb 1 when the switch S is closed?

B

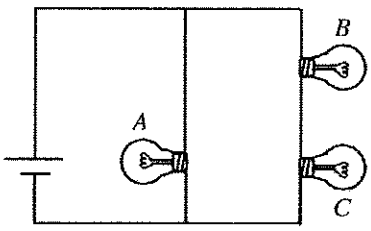
more current
Less R



A) The brightness will increase momentarily then return to its previous level.
B) The brightness increases permanently.
C) The brightness will decrease momentarily then return to its previous level.
D) The brightness remains the same as before the switch is closed.
E) The brightness decreases permanently.

3) In the circuit shown in the figure, all the lightbulbs are identical. Which of the following is the correct ranking of the brightness of the bulbs?

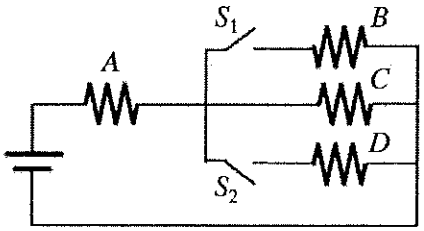
D



A) B and C have equal brightness, and A is the dimmest.
B) A and B have equal brightness, and C is the dimmest.
C) A is brightest, C is dimmest, and B is in between.
D) A is the brightest, and B and C have equal brightness but less than A.
E) All three bulbs have the same brightness.

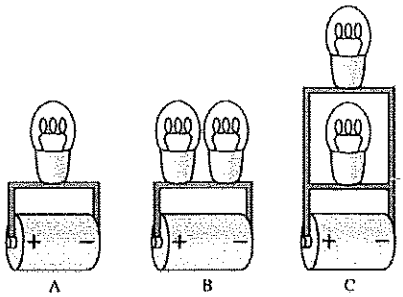
4) In the circuit shown in the figure, four identical resistors labeled A to D are connected to a battery as shown. S_1 and S_2 are switches. Which of the following actions would result in the GREATEST amount of current through resistor A?

A



A) closing both switches
B) closing S_1 only
C) closing S_2 only
D) leaving both switches open as shown.

5) Identical light bulbs can be attached to ideal batteries in three different ways (A, B, or C), as shown in the figure.



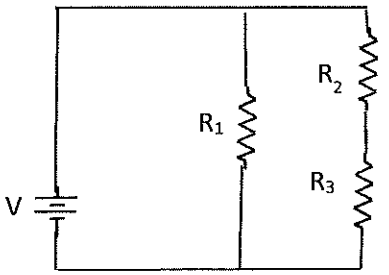
A) If all the batteries are identical, which circuit will require the most current from the battery, rank from highest to lowest or equal?

C > A > B

B) If the light bulbs are all the same brightness, which circuit will require the most potential (voltage) from the battery, rank from highest to lowest or equal?

B > (A = C)

6) Calculate the amount of current through resistors 1 (R_1) and 2 (R_2) in the circuit below.



V = 120 V
R_1 = 600 Ω
R_2 = 200 Ω
R_3 = 100 Ω

$R_{23} = R_2 + R_3 = 300\,\Omega$

$R_{123} = \left(\frac{1}{R_1} + \frac{1}{R_{23}}\right)^{-1} = 200\,\Omega$

$I_1 = \frac{V}{R_1} = 0.2\,A$ | $I_2 = \frac{V}{R_{23}} = 0.4\,A$

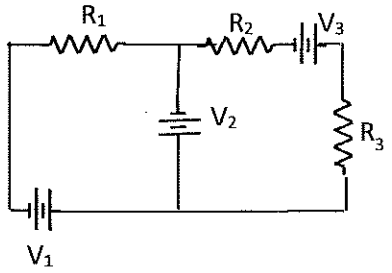
7) Use Kirchhoff's rules to solve for the current through each of the resistors R_1, R_2 and V_2.

See Attached

$I_1 = 0.5\,A$

$I_2 = 0.6\,A$

$I_3 = 1.1\,A$



V_1 = 150 V
V_2 = 200 V
V_3 = 50 V
R_1 = 100 Ω
R_2 = 200 Ω
R_3 = 50 Ω

8) A 12 V dc power supply is connected to two capacitors $C_1 = 10\ \mu F$ and $C_2 = 5\ \mu F$.

A) If the capacitors are wired in series with the power supply determine the

i) charge on each capacitor.

$C_{eq} = \left(\frac{1}{C_1} + \frac{1}{C_2}\right)^{-1} = 3.33\,\mu F$ | $q = 4 \times 10^{-5}\,C$

ii) voltage drop across each capacitor.

$q = C_{eq} V = 4 \times 10^{-5}\,C$ | $V_1 = \frac{q}{C_1} = 4V$ | $V_2 = \frac{q}{C_2} = 8V$

B) If the capacitors are wired in parallel with the power supply determine the

iii) total charge.

$C_{eq} = C_1 + C_2 = 15\,\mu F$ | $q = C_{eq} V = 1.8 \times 10^{-4}\,C$

iv) the charge on each capacitor.

$q_1 = C_1 V = 1.2 \times 10^{-4}\,C$ $\quad q_2 = C_2 V = 6 \times 10^{-5}\,C$

9) A 100 V power supply is wired in series with a 60 μF uncharged capacitor, a switch and a 100 kΩ resistor.

$\tau = RC = 6\,s$ $\quad q_0 = CV = 0.006\,C$

A) When the switch is closed and the capacitor starts to charge, determine the amount of charge that is on the capacitor after t = 10 seconds.

$q(t) = q_0(1 - e^{-t/\tau}) = 0.0049\,C$

B) Calculate the voltage drop across the resistor after 10 s of charging.

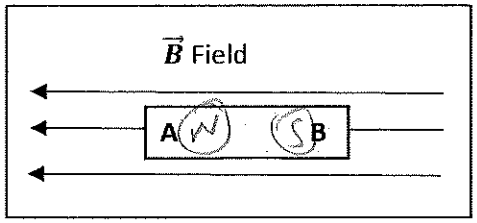
$V(t) = V_0(1 - e^{-t/\tau}) = 81.1\,V$ (capacitor) $\quad | \quad V_R = V_T - V_C = 18.9\,V$ (resistor)

C) How much time will it take to charge the capacitor to a voltage of 20 V?

$t = -\tau \ln\left(\frac{V_0 - V}{V_0}\right) = 8.3\,s$

10) For the permanent magnetic that is free to move in the magnetic field, which poles are represented by A and B?

(A) A is North and B is South.
B) A is South and B is North.
C) A is North and B could be either North or South.
D) A is South and B is North and B could be either North or South.
E) Can not be determined.



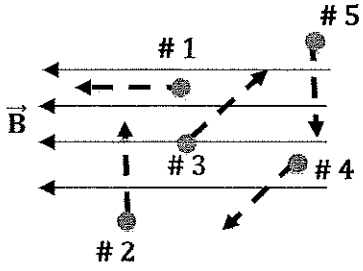
11) An iron nail is more strongly attracted to the

A) north pole of a magnet.
B) south pole of a magnet.
(C) a magnet with a positive charge.
D) a magnet with a negative charge.
E) both the north and south pole of the magnet equally.
F) All the above

12) Five particles with the same net positive charge and velocity are moving in a uniform magnetic field as shown, (assume all particles are in the uniform magnetic field)

A) Rank the 5 charged particles from highest to lowest or equal according to the magnitude of the magnetic force they will experience?

Highest force (2 = 5) (4 = 3) 1 Lowest force (0W)



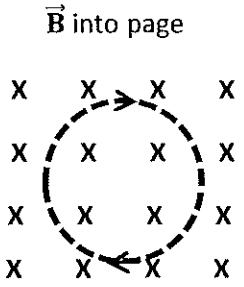
B) Indicate in the space below the direction of the force for each particle, if the force is out of the page use a dot •, if the force is into the page use an X, if to the left, to the right or zero.

#1 no (direct.) #2 • out of page #3 • out of page #4 ⊗ into page #5 ⊗ into page

13) A charged particle is moving in a circular orbit in a uniform magnetic field of $|\vec{B}| = 0.4$ T that is directed into the page, the particle is moving perpendicular to the field.

A) What type of charge must the particle have in order to have a circular path in the direction indicated?

negative charge by RHR #1



B) What is the mass of the particle if it has a charge of q = 400 μC, a speed of $|\vec{v}| = 100$ m/s, and moves in a circular path of radius r = 1.5 m?

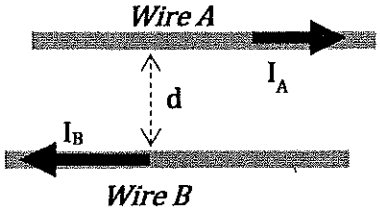
$F_m = ma_c$ $\quad UBq = \frac{mv^2}{r}$ $\quad \rightarrow m = \frac{rBq}{v} = 2.4 \times 10^{-6}$ kg

14) Two long straight wires both have a current in the directions shown. The wires are L = 0.5 m long and are separated by d = 5 mm.

A) Determine the direction of the magnetic force on each wire.

Wire A: up $+\hat{y}$

Wire B: down $-\hat{y}$



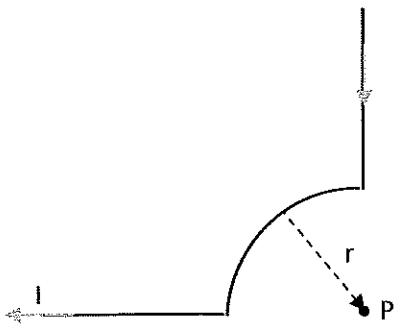
B) Determine the magnetic force that wire A exerts on wire B.

Wire A with $I_A = 10$ A Wire B with $I_B = 15$ A

$|\vec{B}_{A\ at\ B}| = \frac{\mu_0 I_A}{2\pi d} = 4 \times 10^{-4}$ T

$|\vec{F}_{A\ on\ B}| = I_B L |\vec{B}_{A\ at\ B}| = 0.003$ N

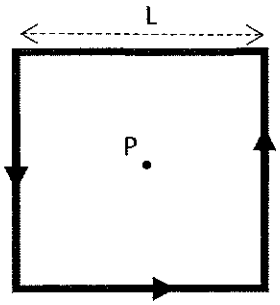
15) Using Biot-Savart Law, derive the equation for the magnetic field at point P, at the center of a quarter loop of current carrying wire. Consider only the loop of wire not the straight end pieces. Label the $d\vec{L}$, $\hat{r}$, direction of the magnetic field and any other variables need to solve the problem.



See attached

$\vec{B} = \left(+\frac{I\mu_0}{8r}\right)\hat{z}$

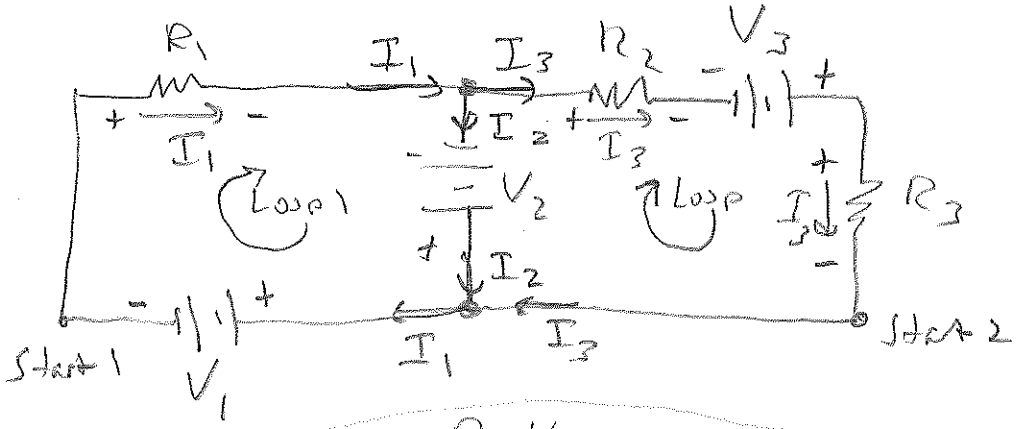
16) Using Biot-Savart Law, derive the equation for the magnetic field at point P, at the center of a square loop of current carrying wire with length L and current flowing counterclockwise. Label the $d\vec{L}$, $\hat{r}$, direction of the magnetic field and any other variables need to solve the problem.



see attached

$$\vec{B} = \left[4\left(\frac{\mu_0 I}{a\pi\sqrt{2}}\right)\right]\hat{z}$$

7)



Pick current direction

Junction rule $I_1 = I_2 + I_3$

- Pick loop + direction of loop
- Indicate "+" and "−" For batteries
- define "+" and "−" for resistor by defined current direction from your Junctions

Loop 1

$$(-I_1 R_1) + (V_2) + (-V_1) = 0$$

$$I_1 = \frac{V_2 - V_1}{R_1} = 0.5A$$

Loop 2

$$(-V_2) + (-I_3 R_2) + V_3 + (-I_3 R_3) = 0$$

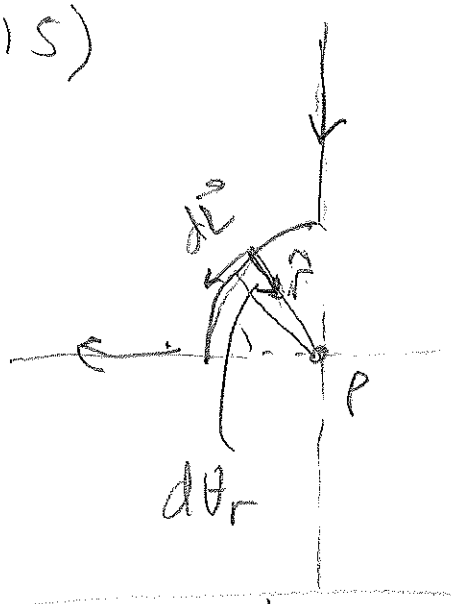
$$I_3 = \frac{V_3 - V_2}{(R_2 + R_3)} = -0.6A$$

$I_1 = I_2 + I_3$

$I_2 = I_1 - I_3 = 0.5A - (-0.6A)$

$(I_2 = 1.1A)$

15)



$$\vec{B} = \frac{\mu_0}{4\pi}\int \frac{I\, d\vec{L}\times\hat{r}}{r^2}$$

$$\vec{B}\cdot\hat{z} = |\vec{B}|\cos\phi \qquad \phi = 0^\circ$$

$$B_z = +|\vec{B}| \quad \text{along } \hat{z} \text{ axis}$$

$$d\vec{L}\times\hat{r} = |d\vec{L}|\sin\phi$$

angle between $d\vec{L}$ and $\hat{r}$

$$\phi = 90^\circ$$

$$|d\vec{L}\times\hat{r}| = |d\vec{L}|$$

$$d\theta_r = \frac{dL}{r}$$

$$dL = r\, d\theta_r$$

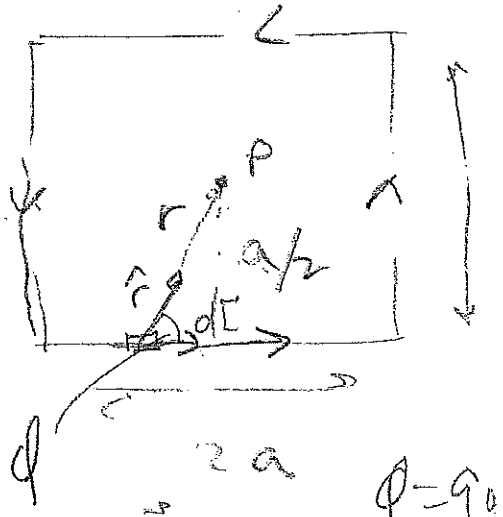
$$|\vec{B}| = \frac{I\mu_0}{4\pi}\int_{\pi/2}^{\pi}\frac{dL}{r^2} = \frac{I\mu_0}{4\pi r^2}\int_{\pi/2}^{\pi} r\, d\theta_r$$

$$|\vec{B}| = \frac{I\mu_0}{4\pi r}\int_{\pi/2}^{\pi} d\theta_r = \frac{I\mu_0}{4\pi r}\left[\theta_r\right]_{\pi/2}^{\pi} \ldots$$

$$|\vec{B}| = \frac{I\mu_0}{8r}$$

$$\vec{B} = \left(+\frac{I\mu_0}{8r}\right)\hat{z}$$

16) Biot - Savart



$d\vec{L} \times \hat{r} = dL \sin\phi$ (dx ↙, $\sin\phi$ ↙)

$\sin\phi = \frac{(a/2)}{r}$

$r = ((a/2)^2 + x^2)^{1/2}$

$\vec{B}$ out of page

$\vec{B} \cdot \hat{L} = |\vec{B}| \sin\phi = |\vec{B}|$

$$B_z = \frac{\mu_0}{4\pi} \int \frac{I\, d\vec{L} \times \hat{r}}{r^2} = \frac{\mu_0 I}{4\pi} \int_{-a/2}^{a/2} \frac{\sin\phi}{r^2}\, dx$$

$$\frac{\mu_0 I}{4\pi} \int_{-a/2}^{a/2} \left(\frac{(a/2)}{((a/2)^2 + x^2)^{1/2}} \right) \left(\frac{1}{((a/2)^2 + x^2)} \right) dx$$

$$= \frac{\mu_0 I}{4\pi} \int_{-a/2}^{a/2} \frac{(a/2)}{((a/2)^2 + x^2)^{3/2}}\, dx = \frac{\mu_0 I (a/2)}{4\pi} \left[\frac{x}{(a/2)^2 (x^2 + (a/2)^2)^{1/2}} \right]_{-a/2}^{+a/2}$$

$$= \frac{\mu_0 I}{4\pi (a/2)} \left[\frac{(a/2)}{((a/2)^2 + (a/2)^2)^{1/2}} - \frac{(-a/2)}{((a/2)^2 + (a/2)^2)^{1/2}} \right]$$

$$= \frac{\mu_0 I}{4\pi (a/2)} \left[2 \left(\frac{a/2}{((a/2)^2 + (a/2)^2)^{1/2}} \right) \right]$$

$$= \frac{\mu_0 I}{2\pi a} \left[\frac{a}{(2(a/2)^2)^{1/2}} \right]$$

$$= \frac{\mu_0 I}{2\pi} \left[\frac{1}{\sqrt{2}(a/2)} \right]$$

$$B_z = \frac{\mu_0 I}{a\pi\sqrt{2}}$$

For all 4 - wires

$$B_z = 4 \left(\frac{\mu_0 I}{a\pi\sqrt{2}} \right)$$